

REFLECTED LAYERS AND A CUBE-ROOT SAVING FOR DISTINCT SUBSET SUMS

ABSTRACT. Let $f(n)$ be the least possible maximum of a set of n positive integers with distinct subset sums. We give a construction proving $f(n) \leq ((9/4)^{1/3} + o(1))2^n/n^{1/3}$. A directed graph arranged in overlapping reflected layers gives modular labels whose equal-cardinality subset sums are distinct. Its normalized determinant is a return probability of a finite reversible Markov chain. An explicit Green function bounds that return probability. Binary padding then converts the modular labels to positive integers.

1. STATEMENT AND MODULAR REDUCTION

A finite set of positive integers is *sum-distinct* if all its subset sums, including the empty sum, are distinct. Define

$$f(n) = \min\{\max A : |A| = n, A \text{ is sum-distinct}\}.$$

Logarithms used in choosing binary padding are to base two.

Theorem 1.1. *As $n \rightarrow \infty$,*

$$f(n) \leq \left(\left(\frac{9}{4} \right)^{1/3} + o(1) \right) \frac{2^n}{n^{1/3}}.$$

In particular, $f(n) = o(2^n / \log_2 n)$.

The following elementary conversion is the same one used for the dyadic construction in `Erdos1.tex`. We include its proof so that the present argument is self-contained.

Lemma 1.2 (Binary padding). *Suppose $C \subseteq \{0, \dots, q-1\}$ has size r and its equal-cardinality subset sums are distinct modulo q . For any integer $t \geq 0$, put*

$$b = (2^t + r)q, \quad A = \{2^i q : 0 \leq i < t\} \cup \{b + c : c \in C\}.$$

Then A is sum-distinct, $|A| = r + t$, and

$$\frac{\max A}{2^{r+t}} < \left(1 + \frac{r+1}{2^t} \right) \frac{q}{2^r}.$$

Proof. The shifted labels exceed every binary element and are pairwise distinct. A collision would give, with $\alpha_i, \beta_c \in \{-1, 0, 1\}$,

$$q \sum_{i < t} \alpha_i 2^i + b \sum_{c \in C} \beta_c + \sum_{c \in C} \beta_c c = 0.$$

The absolute value of the first and third terms together is at most $(2^t - 1)q + r(q - 1) < b$. Thus $\sum_c \beta_c = 0$. Reduction modulo q and the hypothesis on C give $\beta = 0$; uniqueness of binary expansion then gives $\alpha = 0$. Finally, $\max A < (2^t + r + 1)q$. $\square$

2. THE GRAPH AND ITS LATTICE

Fix integers $K \geq 2$ and $s \geq 1$. Write

$$w = K - 1, \quad L = 2s, \quad r = Lw + 1.$$

The vertices are $0, 1, \dots, Lw$, with $a = 0$ and $d = Lw$. For $0 \leq h < L$ and $0 \leq j \leq w$, use the notation

$$v_{h,j} = hw + j.$$

Thus adjacent rows overlap in one vertex: $v_{h,w} = v_{h+1,0}$. For $0 \leq j < w$, give $v_{h,j}$ the two outgoing edges

$$v_{h,j} \longrightarrow v_{h,j+1}, \quad v_{h,j} \longrightarrow \begin{cases} v_{h+1,w-j}, & h < L-1, \\ a, & h = L-1. \end{cases} \quad (2.1)$$

Give d the single edge $d \rightarrow a$. Every vertex except d occurs exactly once as a source in (2.1); the overlapping row endpoints are not additional vertices.

The edges $i \rightarrow i+1$, together with $d \rightarrow a$, form a Hamilton cycle. Every other edge either returns to a or increases the vertex index by $2(w-j) \geq 2$. In particular, deletion of a makes the graph acyclic. Let E be its adjacency matrix, with outgoing edges in rows, and put

$$M = 2I_r - E, \quad q = \det M, \quad \mathcal{L} = \mathbb{Z}^r M.$$

All row sums of M are zero except the row at d , whose sum is one.

Lemma 2.1 (Propagation through the rows). *There is no set S of vertices satisfying all the following conditions:*

- (1) S contains a directed cycle and does not contain d ;
- (2) every vertex in S has a predecessor in S ;
- (3) two distinct predecessors in S force their successor to lie in S .

Proof. Every cycle contains a , and a cycle avoiding d returns to a from the last row. Between its departure and return it follows strictly increasing indices. Such a path meets every closed row interval $[hw, (h+1)w]$: an edge starting below hw cannot end above $(h+1)w$. Consequently S meets every row.

In row zero, every vertex other than a has just its immediately preceding vertex as a predecessor. Condition (2) therefore makes the positions in S in this row an initial interval $\{0, \dots, u\}$.

For a later row $h \geq 1$, position $j \in \{1, \dots, w\}$ has precisely the two distinct predecessors

$$v_{h,j-1} \quad \text{and} \quad v_{h-1,w-j}. \tag{2.2}$$

Position zero is the last vertex of the preceding row. Suppose first that S occupies an initial interval $\{0, \dots, u\}$ in row $h-1$. If $u = w$, condition (3) fills row h . Otherwise its position zero is absent. Its first position j in S must, by condition (2), have its reflected predecessor in S , so $w-j \leq u$. All subsequent positions also have reflected predecessors in S . Applying condition (3) successively fills positions $j, j+1, \dots, w$. Thus row h has a nonempty final interval.

Conversely, suppose row $h-1$ has a final interval $\{\ell, \dots, w\}$. Position zero in row h is in S . Its reflected inputs are in S exactly at positions $1, \dots, w-\ell$, which condition (3) fills. Beyond that initial interval, condition (2) says that a position in S must have its immediately preceding position in S . Hence the positions in S form an initial interval, with no later restart after a gap.

The rows therefore alternate between initial and final intervals. Since L is even, the last row has a final interval and contains d , a contradiction. $\square$

Lemma 2.2 (Balanced ternary avoidance). *If $p \in \mathcal{L}$, all its coordinates belong to $\{-1, 0, 1\}$, and $\sum_j p_j = 0$, then $p = 0$.*

Proof. Write $p = zM$ with $z \in \mathbb{Z}^r$. The row sums give $z_d = \sum_j p_j = 0$. If $z_j > 0$ but no predecessor has positive coefficient, then

$$p_j = 2z_j - \sum_{i \rightarrow j} z_i \geq 2,$$

which is impossible. The analogous assertion holds for negative coefficients. Following predecessors within either nonempty sign class produces a cycle, and every cycle contains a . Both sign classes therefore cannot occur.

If $z \neq 0$, reverse signs if necessary so that $z \geq 0$ and let $S = \{j : z_j > 0\}$. This set contains a cycle, avoids d , and satisfies condition (2) of Lemma 2.1. Two predecessors in S give

$$2z_j = p_j + \sum_{i \rightarrow j} z_i \geq -1 + 2 = 1,$$

so condition (3) also holds. Lemma 2.1 gives a contradiction. Thus $z = 0$ and $p = 0$. $\square$

Lemma 2.3 (Determinant and labels). *The integer q is positive and odd. There are r distinct residues modulo q whose equal-cardinality subset sums are distinct. Moreover, $q/2^r$ is the probability that a walk starting at a , choosing each outgoing edge with probability $1/2$, dies before its first return to a , where the missing probability at d means death.*

Proof. Put $B = E/2$. The submatrix on vertices other than a is strictly upper triangular. Its geometric series gives the total return probability R , and the Schur complement gives

$$\det(I_r - B) = 1 - R = \frac{q}{2^r}.$$

All excursions are finite; following the Hamilton path to d and then dying has positive probability. Thus $q > 0$.

Every cycle cover of E must be a single Hamilton cycle, because every cycle contains a . The increasing order of the remaining vertices makes the displayed Hamilton cycle the only one. Hence $\det E = \pm 1$, and $M \equiv E \pmod{2}$ shows that q is odd.

Let $c = \text{adj}(M)e_a$. Then $Mc = qe_a$, while the principal minor obtained by deleting a is 2^{r-1} , so $c_a = 2^{r-1}$. The homomorphism

$$\phi : \mathbb{Z}^r \longrightarrow \mathbb{Z}/q\mathbb{Z}, \quad p \longmapsto pc \pmod{q}$$

is surjective and vanishes on $\mathcal{L}$. Both its kernel and $\mathcal{L}$ have index q , the latter by the determinant formula for the index of a full-rank integer row lattice. Thus $\ker \phi = \mathcal{L}$. Lemma 2.2 now proves the desired property of the coordinate labels $c_j \pmod{q}$. Applying it to $e_i - e_j$ proves their distinctness. $\square$

3. A RETURN-PROBABILITY ESTIMATE

Number the positions in a row by $0, \dots, K-1$. In a row other than the last, the walk either takes a reflected edge or continues to the shared endpoint, which is position zero of the next row. Its transition matrix between rows is

$$P_{i0} = 2^{i+1-K}, \quad P_{ij} = \begin{cases} 2^{i+j-K}, & 1 \leq j \leq K-1-i, \\ 0, & j > K-1-i. \end{cases} \quad (3.1)$$

In particular, $P_{K-1,0} = 1$: entering the shared endpoint simply enters the next row at position zero, without taking another edge. In the final row, the death probability from position i is $P_{i0}/2$. There are $L = 2s$ rows, so Lemma 2.3 gives

$$\frac{q}{2^r} = \frac{1}{2}(P^{2s})_{00}. \quad (3.2)$$

Proposition 3.1. *For the matrix (3.1) and every integer $s \geq 0$,*

$$\frac{1}{2}(P^{2s})_{00} \leq \frac{1}{K+1} + \frac{K}{6(s+1)}.$$

Proof. The matrix P is stochastic and reversible for

$$\pi_0 = \frac{2}{K+1}, \quad \pi_i = \frac{1}{K+1} \quad (1 \leq i < K).$$

Detailed balance follows immediately from (3.1). Let $Q = P^2$ and $A = Q^{-1} - I_K$. Direct inversion gives

$$\begin{aligned} (Af)_0 &= f_0 - f_1, \\ (Af)_i &= 2(2f_i - f_{i-1} - f_{i+1}) & (1 \leq i < K-1), \\ (Af)_{K-1} &= 2(f_{K-1} - f_{K-2}). \end{aligned}$$

For completeness, write $P = GJ$, where J reverses the coordinate order and G is upper triangular. Then G^{-1} has diagonal $(2, \dots, 2, 1)$ and superdiagonal $(-1, \dots, -1)$. Multiplying $Q^{-1} = JG^{-1}JG^{-1}$ gives the displayed formulas.

The operator A is self-adjoint in $\ell^2(\pi)$, and

$$\langle f, Af \rangle_\pi = \frac{2}{K+1} \sum_{i=0}^{K-2} (f_{i+1} - f_i)^2.$$

Its kernel consists of constants. Thus $Q = (I_K + A)^{-1}$ has eigenvalue 1 on constants and all its other eigenvalues lie in $(0, 1)$. By its orthogonal spectral decomposition, the sequence

$$b_j = (Q^j)_{00} - \pi_0 \quad (j \geq 0)$$

is nonnegative, nonincreasing, and summable.

We compute its sum explicitly. Let $g_i = \mathbf{1}_{i=0} - \pi_0$, and let f be the mean-zero solution of $Af = g$. The coordinate formulas show that

$$f_i - f_{i+1} = \frac{K-1-i}{K+1} \quad (0 \leq i < K-1).$$

Using $\sum_i \pi_i f_i = 0$ gives

$$f_0 = \frac{K(K-1)(2K-1)}{6(K+1)^2}.$$

Since $(I_K - Q)(f + g) = g$, the mean-zero geometric series for Q gives

$$\begin{aligned} \sum_{j=0}^{\infty} b_j &= f_0 + g_0 \\ &= \frac{K(K-1)(2K-1)}{6(K+1)^2} + \frac{K-1}{K+1} \\ &= \frac{K}{3} - \frac{K^2 + K + 6}{6(K+1)^2} < \frac{K}{3}. \end{aligned}$$

Monotonicity therefore gives $(s+1)b_s \leq \sum_{j \geq 0} b_j < K/3$. Combining this with $\pi_0 = 2/(K+1)$ proves the proposition. $\square$

4. CHOOSING THE PARAMETERS

For each integer $K \geq 2$, choose

$$s_K = \left\lceil \frac{K^2}{3} \right\rceil, \quad r_K = 2s_K(K-1) + 1.$$

Lemma 2.3 and Proposition 3.1 give a modular set of size r_K with

$$\frac{q_K}{2^{r_K}} \leq \frac{1}{K+1} + \frac{K}{6(s_K+1)} < \frac{3}{2K}. \quad (4.1)$$

Apply Lemma 1.2 with $t_K = 2\lceil \log_2 r_K \rceil$, and put $n_K = r_K + t_K$. The resulting positive integer set has maximum N_K satisfying

$$\frac{N_K}{2^{n_K}} < (1 + O(r_K^{-1})) \frac{3}{2K}.$$

The integers n_K increase with K , and

$$n_K = \frac{2}{3}K^3 + O(K^2), \quad \frac{n_{K+1}}{n_K} \longrightarrow 1.$$

For an arbitrary sufficiently large n , choose the largest K with $n_K \leq n$. A sum-distinct set A can be enlarged by replacing it with $\{2a : a \in A\} \cup \{1\}$. Parity proves that the new set is sum-distinct; its maximum doubles, so its maximum divided by $2^{|A|}$ is unchanged. Apply this enlargement $n - n_K$ times. Since $K = (3n/2)^{1/3}(1 + o(1))$, we obtain

$$\frac{f(n)}{2^n} \leq \frac{3}{2K}(1 + o(1)) = \left(\left(\frac{9}{4} \right)^{1/3} + o(1) \right) n^{-1/3}.$$

This proves Theorem 1.1. The argument establishes a cube-root saving; it does not establish an upper bound of order $2^n/\sqrt{n}$.